VehicleSure

System Workflow Document

WillowShield – Spring Boot Backend

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Base URL: /api

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1. System Overview & Roles

VehicleSure (codebase: WillowShield) is a Spring Boot vehicle insurance management platform with an Angular frontend. It manages the complete vehicle insurance lifecycle from quote generation to claim settlement.

1.1 System Roles

Role	Description
CUSTOMER	Public user who registers, buys policies, and submits claims
ADMIN	Platform administrator; manages staff, plans, applications, claims, and payouts
UNDERWRITER	Reviews and approves or rejects assigned vehicle applications
CLAIMS_OFFICER	Reviews and approves or rejects assigned insurance claims

1.2 Technology Stack

Component	Technology
Backend	Spring Boot (Java) – WillowShield
Frontend	Angular
Database	MySQL
Security	JWT (JSON Web Tokens) + BCrypt password encoding
Payments	Razorpay (premium collection) + RazorpayX (claim payouts)
Email	n8n Webhook (staff onboarding emails)
Documents	PDF generation for invoices & policy documents

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2. Workflow 1 – Customer Registration

Actors: Anonymous User (Guest)

2.1 Process Steps

#	Step	Description
1	Submit Registration Form	Customer provides name, email, password, and phone number
2	Input Validation	Backend checks required fields, email format, password ≥ 8 chars, phone regex
3	Duplicate Check	Email and phone number uniqueness verified in DB
4	Password Encoding	Password encoded using BCrypt (passwordEncoder.encode)
5	User Record Created	User saved to my_user table with role=CUSTOMER and isActive=true
6	JWT Token Generated	Token contains userId, username, role
7	Response Returned	JWT token + username + role returned to client

2.2 API Endpoint

```
POST /api/auth/register
```

2.3 Request Fields (RegisterReqDTO)

Field	Type	Validation
username	String	@NotBlank, 3–100 characters
password	String	@NotBlank, 8–50 characters
email	String	@NotBlank, valid email format
phoneNumber	String	@NotBlank, 10–15 digits, may include + country code

2.4 Response Fields (RegisterResDTO)

Field	Value
token	String (JWT)
username	String
role	String – always CUSTOMER

2.5 Business Rules & Decision Points

Rule	Behaviour
Duplicate email	Throws DuplicateResourceException → 400 Bad Request

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Rule	Behaviour
Duplicate phone	Throws DuplicateResourceException → 400 Bad Request
Default role	Always assigned CUSTOMER on registration
Phone regex	^\+?[0-9]{10,15}$

Notifications: None triggered on registration.

Outcome: Customer account created. JWT token returned for immediate authentication.

3. Workflow 2 – Login / Authentication

Actors: Any registered user (CUSTOMER, ADMIN, UNDERWRITER, CLAIMS_OFFICER)

3.1 Process Steps

#	Step	Description
1	Submit Credentials	User sends email + password to login endpoint
2	Spring Security Authenticate	AuthenticationManager.authenticate() called
3	Fetch User Entity	User retrieved from DB by email via userRepository.findByEmail()
4	Generate JWT	Token contains userId (subject), username, role
5	Return Token	Client stores JWT and sends as Authorization: Bearer <token> on all requests

3.2 API Endpoint

```
POST /api/auth/login
```

3.3 Request / Response

Request Field (JwtRequest)	Description
email	String – user's registered email
password	String – plain text (transmitted over HTTPS)

Response Field (JwtResponse)	Value
token	JWT String – used in all subsequent API calls

Decision Points: Invalid credentials → Spring Security throws BadCredentialsException.

4. Workflow 3 – Quote Generation

Actors: Customer (authenticated)

4.1 Process Steps

#	Step	Description
1	Submit Vehicle Details	Customer sends ex-showroom price, vehicle type, year, accidents, distance driven
2	Calculate Vehicle Age	vehicleAge = currentYear - year
3	Determine Risk Level	Based on age, accidents, and distance driven (see Business Rules)
4	Calculate IDV	IDV = exShowroomPrice × (1 - depreciationRate), max 50% depreciation
5	IDV Multiplier	Bracket-based multiplier applied to adjust premium for vehicle value
6	Risk Multiplier	LOW=1.0, MEDIUM=1.2, HIGH=1.5
7	Fetch Active Plans	Active PolicyPlans filtered by vehicleType
8	Calculate Premiums	Computed for 1-year, 2-year, and 3-year tenures per plan
9	Return Response	List of plans with quote options for each tenure

4.2 API Endpoint

```
POST /api/customer/quote
```

4.3 Business Rules & Formulas

Risk Level Calculation

```
riskScore = 0
if (vehicleAge > 5):          riskScore += 1   // Old vehicle
riskScore += accidentsInPast                    // Each accident adds 1
if (distanceDriven > 100000): riskScore += 1   // High mileage
LOW = riskScore <= 1  |  MEDIUM = riskScore == 2  |  HIGH = riskScore >= 3
```

IDV Calculation

```
depreciation = 10% × vehicleAge  (capped at 50%)
IDV = exShowroomPrice × (1 - depreciation)
```

IDV Multiplier Brackets

IDV Range	Multiplier	Effect
IDV > ₹10,00,000	1.3	+30% premium adjustment
IDV > ₹5,00,000	1.1	+10% premium adjustment
IDV < ₹2,00,000	0.9	-10% premium adjustment

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IDV Range	Multiplier	Effect
Otherwise	1.0	No adjustment

Premium Calculation by Tenure

```
finalMultiplier = riskMultiplier × idvMultiplier
1-Year  = basePremium × finalMultiplier × 1.0  (no discount)
2-Year  = basePremium × finalMultiplier × 1.9  (5% multi-year discount)
3-Year  = basePremium × finalMultiplier × 2.8  (6.7% multi-year discount)
```

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5. Workflow 4 – Policy Purchase (Application Submission)

Actors: Customer (authenticated)

5.1 Process Steps

#	Step	Description
1	Submit Multipart Form	JSON application data + RC document + Invoice document
2	Extract Customer Identity	customerId from JWT in security context
3	Validate Customer	Must exist in DB
4	Validate Policy Plan	Plan must exist and have isActive = true
5	Vehicle Type Compatibility	Plan's applicableVehicleType must match submitted vehicleType
6	Validate Registration Number	Format: XX00XX0000 (Indian pattern)
7	Validate Chassis Number	17-char VIN (no I, O, Q allowed)
8	Validate Year	Manufacturing year cannot be in the future
9	Validate Distance	Cannot be negative
10	Duplicate Application Check	No non-REJECTED application with same registration number
11	Calculate Risk Level	Using risk formula (see Workflow 3)
12	Create VehicleApplication	Status set to UNDER_REVIEW
13	Save Documents to Disk	RC: uploads/documents/app_<ID>_RC.<ext> Invoice: app_<ID>_Invoice.<ext>
14	Create VehicleDocument	RC and invoice paths stored in DB
15	Notify Admins	In-app notification to all admin users

5.2 API Endpoint

```
POST /api/customer/buy-policy
Content-Type: multipart/form-data
```

5.3 Request Parts

Part	Description
application	String (JSON) – CreateVehicleApplicationDTO
rcDocument	MultipartFile – RC book scan or PDF

Part	Description
invoiceDocument	MultipartFile – Vehicle purchase invoice

5.4 Application DTO Fields

Field	Type	Validation
vehicleOwnerName	String	@NotBlank
registrationNumber	String	@NotBlank, format XX00XX0000
make	String	@NotBlank
model	String	@NotBlank
year	Integer	@NotNull, ≤ currentYear
fuelType	String	@NotBlank
chassisNumber	String	@NotBlank, 17-char VIN
distanceDriven	Long	@NotNull, ≥ 0
exShowroomPrice	BigDecimal	@NotNull, ≥ 0
idv	BigDecimal	@NotNull (from quote)
calculatedPremium	BigDecimal	@NotNull (selected from quote)
tenureYears	Integer	@NotNull, ≥ 1
vehicleType	Enum	@NotNull
planId	Integer	@NotNull
transmissionType	Enum	@NotNull
accidentsInPast	Integer	@NotNull, ≥ 0

5.5 Error Decision Points

Condition	Exception Thrown
Plan not active	InvalidStateException
Vehicle type mismatch	InvalidStateException
Invalid reg number format	BusinessException
Invalid chassis number	BusinessException
Year in future	BusinessException
Negative distance	BusinessException
Duplicate active application	DuplicateResourceException

Notification: Admin in-app – "New policy application submitted by <customer> for vehicle <regNo>" | Event: NEW_APPLICATION_SUBMITTED

Outcome: VehicleApplication created with status UNDER_REVIEW. Admin notified.

6. Workflow 5 – Underwriting Review

Actors: Admin (assigns), Underwriter (reviews and decides)

Phase A – Admin Assigns Application

#	Step	Description
1	Admin Views All Applications	GET /api/admin/applications
2	Admin Views Underwriter Workload	Count of currently ASSIGNED applications per underwriter
3	Admin Assigns to Underwriter	Application status → ASSIGNED
4	Underwriter Notified	In-app notification sent to assigned underwriter

Phase A – API Endpoints

Endpoint	Purpose
GET /api/admin/applications	View all vehicle applications
GET /api/admin/underwriters/workload	View ASSIGNED count per underwriter
PUT /api/admin/applications/{id}/assign/{uwId}	Assign application to underwriter

Validation: Application must be in UNDER_REVIEW status. Already ASSIGNED/APPROVED → InvalidStateException.

Phase B – Underwriter Reviews & Decides

#	Step	Description
1	View Assigned Applications	Filtered by underwriter's ID only
2	Review Details	Vehicle details, chassis, RC document, invoice document
3	Make Decision	Approve or Reject the application

On APPROVAL – Actions Taken

#	Action
1	Status validated as ASSIGNED or UNDER_REVIEW
2	Vehicle entity created from application data
3	Policy entity created with status PENDING_PAYMENT
4	Policy number generated: POL-<8-char UUID>
5	Policy dates set: startDate = today, endDate = today + tenureYears

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#	Action
6	Application status → APPROVED
7	Customer in-app notification sent

On REJECTION – Actions Taken

#	Action
1	Status validated as ASSIGNED or UNDER_REVIEW
2	Application status → REJECTED
3	Rejection reason stored on application record
4	Customer in-app notification sent with reason

Phase B – API Endpoints

Endpoint	Purpose
GET /api/underwriter/applications	View only assigned applications
POST /api/underwriter/applications/{id}/approve	Approve an assigned application
POST /api/underwriter/applications/{id}/reject	Reject with reason: { reason: string }

Event	Notification
Approval	Customer: "Your application for <regNo> has been approved. Proceed with payment." | Event: POLICY_APPROVED
Rejection	Customer: "Your application for <regNo> has been rejected. Reason: <reason>" | Event: POLICY_REJECTED

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7. Workflow 6 – Premium Payment (Razorpay)

Actors: Customer (authenticated)
There are two payment flows – the recommended Razorpay flow and a legacy direct payment option.

Flow A – Razorpay Payment (Recommended)

#	Step	Description
1	Customer Initiates Payment	Selects approved policy and clicks Pay
2	Create Razorpay Order	POST /api/payments/create-order – Backend calls razorpayClient.orders.create()
3	Order ID Returned	Razorpay orderId + currency + keyId returned to frontend
4	Customer Completes Checkout	Customer completes payment on Razorpay checkout widget
5	Frontend Sends Verification	POST /api/payments/verify with razorpayOrderId, paymentId, signature
6	Backend Verifies Signature	HMAC-SHA256 signature verified via Utils.verifyPaymentSignature()
7	Finalize Payment	If valid: payment recorded, PDFs generated, policy activated

On Successful Verification – Actions

#	Action
1	Payment record created: status=PAID, transactionRef=razorpayPaymentId
2	VehicleApplication status → PAID
3	Invoice PDF generated: uploads/invoices/policy_<id>_invoice_<nano>.pdf
4	Policy Document PDF generated: uploads/policies/policy_<id>_document_<nano>.pdf
5	Policy status → ACTIVE
6	Customer in-app notification sent
7	Admin in-app notification sent

Razorpay API Endpoints

Endpoint	Purpose
POST /api/payments/create-order	Create Razorpay order (amount in paise = amount × 100)
POST /api/payments/verify	Verify HMAC signature and finalize payment
GET /api/payments/download/invoice/{policyId}	Download Invoice PDF

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Endpoint	Purpose
GET /api/payments/download/policy/{policyId}	Download Policy Document PDF

Flow B – Legacy Direct Payment

Property	Detail
Endpoint	POST /api/payments/process
Extra Field	transactionReference: String (manual entry)
Verification	No Razorpay signature check – direct processing
DB Actions	Same payment record + PDF generation + policy activation
Security	Only the policy owner can pay (validated from JWT)

Recipient	Notification
Customer	"Premium payment for policy <policyNo> successful. Policy is now active." | Event: PAYMENT_SUCCESS
Admin	"Premium paid via Razorpay for policy <policyNo> by <customer>" | Event: PREMIUM_PAID

Outcome: Policy is ACTIVE. Invoice and Policy Document PDFs available for download.

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8. Workflow 7 – Claim Submission

Actors: Customer (authenticated)

8.1 Process Steps

#	Step	Description
1	Customer Selects Policy	Must be an active policy owned by the customer
2	Fill Claim Form	Claim type, description, bank account details
3	Upload Documents	Up to 3 supporting files (photos, repair estimates, etc.)
4	Validate No Active Claim	No SUBMITTED/ASSIGNED/APPROVED claim exists for this policy
5	Create Claim Entity	Status = SUBMITTED, claimNumber = CLM-<8-char UUID>
6	Store Bank Details	Account number, IFSC code, account holder name stored on claim
7	Save Documents	uploads/claims/claim_<id>_DOC1.<ext> (up to 3 files)
8	Create ClaimDocument Record	Document paths saved in DB
9	Notify Admins	In-app notification to all admins

8.2 API Endpoint

```
POST /api/customer/submit-claim
Content-Type: multipart/form-data
```

8.3 Claim DTO Fields (CreateClaimDTO)

Field	Type	Validation
policyId	Integer	@NotNull
claimType	Enum	@NotNull – DAMAGE, THEFT, THIRD_PARTY
description	String	Optional
bankAccountNumber	String	Optional
ifscCode	String	Optional
accountHolderName	String	Optional

Business Rule: Only one active claim per policy at a time. Duplicate → InvalidStateException.

Notification: Admin in-app – "New claim <claimNo> submitted by <customer>" | Event: NEW_CLAIM_SUBMITTED

Outcome: Claim created at SUBMITTED status. Admin notified to assign to a Claims Officer.

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9. Workflow 8 – Claims Processing

Actors: Admin (assigns), Claims Officer (reviews and decides)

Phase A – Admin Assigns Claim

#	Step	Description
1	Admin Views All Claims	GET /api/admin/claims
2	Admin Views Officer Workloads	Count of ASSIGNED claims per officer
3	Admin Assigns Claim to Officer	Claim status → ASSIGNED
4	Officer Notified	In-app notification to assigned claims officer

Phase A – API Endpoints

Endpoint	Purpose
GET /api/admin/claims	View all claims
GET /api/admin/claims-officers/workload	Count of ASSIGNED claims per officer
PUT /api/admin/claims/{id}/assign/{officerId}	Assign claim (must be SUBMITTED status)

Phase B – Claims Officer Reviews & Decides

#	Step	Description
1	View Assigned Claims	Filtered by officer's ID
2	Review Claim & Documents	Claim details, uploaded evidence
3	Preview Payout (Optional)	POST /api/claims-officer/claims/{id}/payout – calculate before approving
4	Make Decision	Approve or Reject

Payout Preview Request (ApproveClaimRequest)

Field	Type	Description
billAmount	BigDecimal	Repair or damage bill amount
exShowroomPrice	BigDecimal	Vehicle's original ex-showroom price
yearOfManufacture	Integer	Vehicle manufacturing year

On APPROVAL – Actions

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#	Action
1	Payout amount calculated using depreciation formula
2	Claim approvedAmount set
3	Claim status → APPROVED
4	If total-loss scenario: Policy status → INACTIVE
5	Customer in-app notification sent
6	Admin in-app notification sent (payout action required)

On REJECTION – Actions

#	Action
1	Claim status → REJECTED
2	Rejection reason stored
3	Customer in-app notification sent with reason

Event	Notification
Approval (Customer)	"Your claim <claimNo> has been approved for amount: <amount>" | Event: CLAIM_APPROVED
Approval (Admin)	"Claim <claimNo> approved by Claims Officer. Payout: <amount>. Admin action required." | Event: CLAIM_PAYMENT_PENDING
Rejection	"Your claim <claimNo> has been rejected. Reason: <reason>" | Event: CLAIM_REJECTED

Outcome: Claim is APPROVED with calculated payout. Admin notified to process payment.

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10. Workflow 9 – Claim Payout (RazorpayX)

Actors: Admin (authenticated)

10.1 Process Steps

#	Step	Description
1	Admin Views Pending Payouts	GET /api/admin/claims/pending-payments – Claims in APPROVED with no payout yet
2	Admin Triggers Payout	POST /api/admin/claims/{id}/pay
3	Validate Claim Status	Claim must be in APPROVED status
4	RazorpayX Payout Initiated	Bank transfer to customer's account stored on claim
5	ClaimsPayment Record Created	Transaction reference, bank details, status saved to DB
6	Claim Status → SETTLED	Claim closed
7	Customer Notified	In-app notification with settlement confirmation

10.2 API Endpoints

Endpoint	Purpose
GET /api/admin/claims/pending-payments	View approved claims awaiting payout
POST /api/admin/claims/{id}/pay	Trigger RazorpayX bank transfer
GET /api/admin/claims/payouts	View all claim payout history
GET /api/admin/export/payout-history	Download payout history as Excel

10.3 RazorpayX Behaviour

Mode	Behaviour
Production mode (env configured)	Real bank transfer: Create Contact → Create Fund Account → Create Payout
Test mode (env not set)	Returns mock reference: PC_MOCK_<timestamp>
Payout failure	BusinessException thrown; payment record saved with FAILED status

10.4 ClaimsPayment Record Stored

Field	Value
amountPaid	claim.approvedAmount

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Field	Value
bankAccountNumber	Stored from claim
ifscCode	Stored from claim
accountHolderName	Stored from claim
transactionReference	RazorpayX payoutId or mock ID
paymentStatus	COMPLETED or FAILED

Notification: Customer in-app – "Your claim <claimNo> has been settled. Amount: <amount>" | Event: CLAIM_SETTLED

Outcome: Claim is SETTLED. Customer's bank account credited. Full audit trail stored.

11. Workflow 10 – Staff Management (Admin)

11.1 Create Staff

#	Step	Description
1	Admin Provides Details	Username, email, role (UNDERWRITER or CLAIMS_OFFICER), full name, phone
2	Password Handling	If no password provided, system auto-generates secure 10-char password
3	User Created	isActive = true, role set to provided role
4	Welcome Email via n8n	Credentials emailed to staff member via n8n webhook

```
POST /api/admin/staff
```

n8n Email: Subject: "Welcome to VehicleSure - Your Staff Credentials" – contains role, username, and password.
Note: Email failure is logged but does NOT roll back staff creation.

11.2 Staff Management Endpoints

Endpoint	Purpose
POST /api/admin/staff	Create new underwriter or claims officer
DELETE /api/admin/staff/{id}	Deactivate staff (isActive = false)
PUT /api/admin/staff/{id}/activate	Reactivate staff (isActive = true)
GET /api/admin/staff	List all staff
GET /api/admin/staff/{id}	Get staff member by ID

12. Workflow 11 – Policy Plan Management (Admin)

12.1 Process & Endpoints

#	Step	Description
1	Admin Creates Plan	All coverage details, base premium, vehicle type, deductible set
2	Plan Activated	isActive = true by default – immediately visible to customers
3	Admin Deactivates Plan	Plan hidden from customer quote and buy flows
4	Admin Reactivates Plan	Plan made available again

Endpoint	Purpose
POST /api/admin/policy-plans	Create new plan

Endpoint	Purpose
GET /api/admin/policy-plans	List all plans (admin view)
GET /api/admin/policy-plans/{id}	Get plan by ID
DELETE /api/admin/policy-plans/{id}	Deactivate plan
PUT /api/admin/policy-plans/{id}/activate	Activate plan
GET /api/customer/policy-plans	Active plans only (filterable by ?type=CAR)

13. Workflow 12 – Change Password

Actors: Any authenticated user (all roles)

```
PUT /api/auth/change-password
```

#	Step	Description
1	Extract User from JWT	userId from security context
2	Fetch User from DB	User entity loaded
3	Verify Current Password	BCrypt match against stored hash
4	Validate New Password	Cannot be the same as current password
5	Encode & Save	New password BCrypt-encoded and persisted

Condition	Error
Wrong current password	BusinessException: "Current password is incorrect"
Same as current	BusinessException: "New password cannot be same as current password"

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14. Status Transition State Machines

14.1 VehicleApplication Status

```
UNDER_REVIEW  —(Admin assigns)————————————————→ ASSIGNED
UNDER_REVIEW  —(Underwriter approves directly)————→ APPROVED
ASSIGNED      —(Underwriter approves)————————————→ APPROVED
ASSIGNED      —(Underwriter rejects)————————————→ REJECTED
APPROVED      —(Customer pays premium)————————————→ PAID
```

Status	Trigger	Description
UNDER_REVIEW	Application submitted	Awaiting admin assignment to an underwriter
ASSIGNED	Admin assigns to underwriter	Underwriter reviewing the application
APPROVED	Underwriter approves	Policy created, customer can pay premium
REJECTED	Underwriter rejects	Application closed with rejection reason
PAID	Customer pays premium	Policy is now active and live

14.2 Policy Status

```
PENDING_PAYMENT  —(Customer pays)————————————→ ACTIVE
ACTIVE           —(Total-loss claim approved)————→ INACTIVE
ACTIVE           —(End date passed)————————————→ EXPIRED  (logical check)
```

Status	Trigger	Description
DRAFT	Reserved	Not used in current workflows
PENDING_PAYMENT	Policy record created	Premium not yet paid
ACTIVE	Premium paid	Policy is live and valid
INACTIVE	Total-loss claim approved	Policy closed due to total-loss event
EXPIRED	Policy past end date	Coverage ended
CANCELLED	Manual cancellation (reserved)	Reserved for future use

14.3 Claim Status

```
SUBMITTED  —(Admin assigns to officer)————————→ ASSIGNED
ASSIGNED   —(Officer approves)————————————————→ APPROVED
ASSIGNED   —(Officer rejects)————————————————→ REJECTED
APPROVED   —(Admin initiates payout)————————→ SETTLED
```

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Status	Trigger	Description
SUBMITTED	Customer submits claim	Awaiting admin assignment
ASSIGNED	Admin assigns to officer	Claims officer reviewing
APPROVED	Officer approves with amount	Awaiting admin payout processing
REJECTED	Officer rejects with reason	Claim closed, customer notified
SETTLED	Admin processes bank transfer	Claim fully resolved, customer paid

14.4 Payment Status

Status	Description
PAID (PaymentStatus enum)	Premium payment successfully recorded
COMPLETED (ClaimsPayment field)	Claim payout bank transfer successful
FAILED (ClaimsPayment field)	Claim payout bank transfer failed – BusinessException thrown

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15. Business Rules & Calculations

15.1 IDV Depreciation for Claim Payout

Vehicle Age	IDV Depreciation Rate	Formula
≤ 0 years (up to 6 months)	5%	IDV = exShowroomPrice × 0.95
1 year (6m – 1yr)	15%	IDV = exShowroomPrice × 0.85
2 years (1–2yr)	20%	IDV = exShowroomPrice × 0.80
3 years (2–3yr)	30%	IDV = exShowroomPrice × 0.70
4 years (3–4yr)	40%	IDV = exShowroomPrice × 0.60
5 years (4–5yr)	50%	IDV = exShowroomPrice × 0.50
> 5 years	70%	IDV = exShowroomPrice × 0.30

15.2 Parts Depreciation for Claim Payout

Vehicle Age	Parts Depreciation Rate	Formula
≤ 0 years	25%	depreciatedBill = billAmount × 0.75
1 year	30%	depreciatedBill = billAmount × 0.70
2 years	32%	depreciatedBill = billAmount × 0.68
3 years	35%	depreciatedBill = billAmount × 0.65
4 years	38%	depreciatedBill = billAmount × 0.62
5 years	40%	depreciatedBill = billAmount × 0.60
> 5 years	50%	depreciatedBill = billAmount × 0.50

15.3 Claim Payout by Policy Type

Scenario	Calculation Logic
THIRD_PARTY – All claims	Payout = min(depreciatedBill, maxCoverage) - deductible
COMPREHENSIVE – THEFT	Payout = IDV; Policy CLOSED (INACTIVE)
COMPREHENSIVE – DAMAGE (total loss >75% IDV)	Payout = IDV; Policy CLOSED
COMPREHENSIVE – DAMAGE (partial)	Payout = depreciatedBill - deductible
ZERO_DEPRECIATION – THEFT	Payout = IDV; Policy CLOSED
ZERO_DEPRECIATION – DAMAGE (total)	Payout = IDV; Policy CLOSED
ZERO_DEPRECIATION – DAMAGE (partial)	Payout = bill (NO depreciation) - deductible
ZERO_DEPRECIATION – THIRD_PARTY	Payout = min(depreciatedBill, maxCoverage) - deductible

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Scenario	Calculation Logic
Final Rule (all types)	if finalPayout < 0: finalPayout = 0

15.4 Other Business Rules

Rule	Detail
Duplicate Application	Cannot submit new application for a reg number with an existing non-REJECTED application
Duplicate Claim	Cannot submit a new claim if policy has a SUBMITTED/ASSIGNED/APPROVED claim
Plan Eligibility	Plan's applicableVehicleType must match the vehicle being insured
Registration Number	Must match Indian format: ^[A-Z]{2}[0-9]{2}[A-Z]{2}[0-9]{4}$
Chassis (VIN)	Must be 17 chars, no I/O/Q: ^[A-HJ-NPR-Z0-9]{17}$
Manufacturing Year	Cannot be a future year
Distance Driven	Cannot be negative

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16. Role-Based Access Control (RBAC)

16.1 ADMIN – /api/admin/**

Endpoint	Action	Note
POST /api/admin/staff	Create Staff	Create underwriter or claims officer
DELETE /api/admin/staff/{id}	Deactivate Staff	Set isActive = false
PUT /api/admin/staff/{id}/activate	Activate Staff	Set isActive = true
POST /api/admin/policy-plans	Create Policy Plan	
DELETE /api/admin/policy-plans/{id}	Deactivate Plan	
PUT /api/admin/policy-plans/{id}/activate	Activate Plan	
GET /api/admin/applications	View All Applications	
PUT /api/admin/applications/{id}/assign/{uwId}	Assign to Underwriter	Application must be UNDER_REVIEW
GET /api/admin/claims	View All Claims	
PUT /api/admin/claims/{id}/assign/{officerId}	Assign Claim	Claim must be SUBMITTED
GET /api/admin/claims/pending-payments	View Pending Payouts	Approved claims awaiting transfer
POST /api/admin/claims/{id}/pay	Process Payout	RazorpayX bank transfer
GET /api/admin/export/received-payments	Export Payments (Excel)	
GET /api/admin/export/payout-history	Export Payouts (Excel)	

16.2 CUSTOMER – /api/customer/** and /api/payments/**

Endpoint	Action
POST /api/customer/quote	Generate premium quotes
POST /api/customer/buy-policy	Submit vehicle application
GET /api/customer/applications	View own applications only
GET /api/customer/policies	View own active policies
GET /api/customer/policy-plans	View active plans
POST /api/customer/submit-claim	Submit a claim
GET /api/customer/my-claims	View own claims only
POST /api/payments/create-order	Create Razorpay order
POST /api/payments/verify	Verify payment + activate policy
GET /api/payments/download/invoice/{id}	Download invoice PDF

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Endpoint	Action
GET /api/payments/download/policy/{id}	Download policy PDF

16.3 UNDERWRITER – /api/underwriter/**

Endpoint	Action
GET /api/underwriter/applications	View ONLY assigned applications
POST /api/underwriter/applications/{id}/approve	Approve assigned application
POST /api/underwriter/applications/{id}/reject	Reject with reason

16.4 CLAIMS_OFFICER – /api/claims-officer/**

Endpoint	Action
GET /api/claims-officer/claims	View ONLY assigned claims
POST /api/claims-officer/claims/{id}/payout	Preview payout calculation
POST /api/claims-officer/claims/{id}/approve	Approve with payout amount
POST /api/claims-officer/claims/{id}/reject	Reject with reason

16.5 Access Restrictions

Role	Restriction
CUSTOMER	Cannot access other customers' data; no admin/underwriter/officer endpoints
UNDERWRITER	Can only act on applications assigned specifically to them
CLAIMS_OFFICER	Can only act on claims assigned specifically to them
ADMIN	Cannot submit applications or claims; no customer-role endpoints

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17. External Integrations

17.1 Razorpay – Premium Collection

Trigger	Endpoint	What Happens
Customer initiates payment	POST /api/payments/create-order	Backend calls razorpayClient.orders.create() with amount in paise (×100), currency INR
After checkout completion	POST /api/payments/verify	Backend verifies HMAC-SHA256 signature. If valid → payment finalized and policy activated

Config	Detail
Key ID	RAZORPAY_KEY_ID (env variable)
Key Secret	RAZORPAY_KEY_SECRET (env variable)
Amount	Converted to paise: amount × 100
Currency	INR
Receipt	policy_<id>

17.2 RazorpayX – Claim Payouts

Trigger	Endpoint	What Happens
Admin triggers payout	POST /api/admin/claims/{id}/pay	Backend calls razorpayService.createPayout(claim) for bank transfer to customer

Config	Detail
Account Number	RAZORPAYX_ACCOUNT_NUMBER (env variable)
Production Flow	Create Contact → Create Fund Account → Create Payout
Test Mode	If env not set: returns PC_MOCK_<timestamp>
Data Used	bankAccountNumber, ifscCode, accountHolderName, approvedAmount

17.3 n8n – Email Notifications

Trigger	Action	Email Content
Admin creates staff member	POST to n8n webhook URL	"Welcome to VehicleSure" email with role, username, and password sent to new staff

Config	Detail
Webhook URL	n8n.webhook.url (env variable)

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Config	Detail
Payload	{ email, subject, message }
Error Handling	Email failure logged; does NOT roll back the staff creation transaction

18. Notifications Summary

Event	Recipient	Details / Event Type
New application submitted	All Admins	NEW_APPLICATION_SUBMITTED – New application for vehicle <regNo>
Application assigned to UW	Underwriter	NEW_ASSIGNMENT – New application assigned: <regNo>
Application APPROVED	Customer	POLICY_APPROVED – Approved; proceed to pay
Application REJECTED	Customer	POLICY_REJECTED – Rejected: <reason>
Premium payment successful	Customer	PAYMENT_SUCCESS – Policy <no> is now active
Premium payment received	All Admins	PREMIUM_PAID – Premium paid for policy <no>
New claim submitted	All Admins	NEW_CLAIM_SUBMITTED – New claim <claimNo>
Claim assigned to officer	Claims Officer	NEW_ASSIGNMENT – Claim assigned: <claimNo>
Claim APPROVED	Customer	CLAIM_APPROVED – Approved for <amount>
Claim APPROVED (pending pay)	All Admins	CLAIM_PAYMENT_PENDING – Admin payout required
Claim REJECTED	Customer	CLAIM_REJECTED – Rejected: <reason>
Claim SETTLED	Customer	CLAIM_SETTLED – Amount disbursed
Staff created	Staff (Email)	n8n Email – Welcome with login credentials

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